



```
!pip install sentence-transformers
```

```
Requirement already satisfied: sentence-transformers in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: transformers<6.0.0,>=4.41.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: huggingface-hub>=0.20.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typing_extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: packaging>=20.9 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: shellingham in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages
```

```
import re
import numpy as np
import pandas as pd
from sentence_transformers import SentenceTransformer, util

job_description = ""
```

We are hiring a VLSI Engineer with experience in Verilog, FPGA, Digital Design, ASIC flow, Python and Machine Learning basics. Minimum 2 years experience required.

```
"""
resumes = [
    """
    John Doe
    3 years experience in VLSI design.
    Skilled in Verilog, FPGA, ASIC flow and Python.
    """,

    """
    Rahul Sharma
    Software engineer with Python and Machine Learning expertise.
    2 years experience.
    """,

    """
    Priya Singh
    Electronics engineer with FPGA and Digital Design.
    1 year experience.
    """,

    """
    Arjun Mehta
    4 years experience in ASIC design and Verilog.
    Strong background in Digital Systems.
    """,

    """
    Kiran Patel
    Mechanical engineer with CAD and manufacturing skills.
    5 years experience.
    """
]

def clean_text(text):
    text = text.lower()
    text = re.sub(r'\n', ' ', text)
    text = re.sub(r'^a-z0-9\s', '', text)
    return text

def extract_years(text):
    years = re.findall(r'(\d+)\s+year', text)
    if years:
        return int(years[0])
    return 0
```

```
model = SentenceTransformer('all-MiniLM-L6-v2')
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning: The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab. You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models.
warnings.warn(
```

```
modules.json: 0%|          | 0.00/349 [00:00<?, ?B/s]
```

```
config_sentence_transformers.json: 0%|          | 0.00/116 [00:00<?, ?B/s]
```

```
README.md: 0.00B [00:00, ?B/s]
```

```
sentence_bert_config.json: 0%|          | 0.00/53.0 [00:00<?, ?B/s]
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a token as an environment variable (see https://huggingface.co/docs/huggingface_hub/main/en/index#authentication).
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a token as an environment variable (see https://huggingface.co/docs/huggingface_hub/main/en/index#authentication).
```

```
config.json: 0%|          | 0.00/612 [00:00<?, ?B/s]
```

```
model.safetensors: 0%|          | 0.00/90.9M [00:00<?, ?B/s]
```

```
Loading weights: 0%|          | 0/103 [00:00<?, ?it/s]
```

```
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
```

Key	Status	
embeddings.position_ids	UNEXPECTED	

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture

tokenizer_config.json: 0%| | 0.00/350 [00:00<?, ?B/s]

vocab.txt: 0.00B [00:00, ?B/s]

tokenizer.json: 0.00B [00:00, ?B/s]

special_tokens_map.json: 0%| | 0.00/112 [00:00<?, ?B/s]

config.json: 0%| | 0.00/190 [00:00<?, ?B/s]

```
skill_weights = {
    "verilog": 0.2,
    "fpga": 0.2,
    "digital design": 0.2,
    "asic": 0.15,
    "python": 0.15,
    "machine learning": 0.1
}
```

```
def skill_score(text):
    score = 0
    for skill, weight in skill_weights.items():
        if skill in text:
            score += weight
    return score
```

```
def experience_score(years):
    return min(years * 0.1, 1)
```

```
def final_score(semantic, skill, exp):
    return (0.5 * semantic) + (0.3 * skill) + (0.2 * exp)
```

```
jd_clean = clean_text(job_description)
jd_embedding = model.encode(jd_clean, convert_to_tensor=True)
```

```
results = []
```

```
for resume in resumes:
```

```

cleaned = clean_text(resume)
resume_embedding = model.encode(cleaned, convert_to_tensor=True)

semantic = float(util.cos_sim(jd_embedding, resume_embedding))
skill = skill_score(cleaned)
years = extract_years(cleaned)
exp = experience_score(years)

total = final_score(semantic, skill, exp)

name = resume.strip().split("\n")[0]

results.append({
    "Name": name,
    "Semantic Score": semantic,
    "Skill Score": skill,
    "Experience Score": exp,
    "Final AI Fit Score": total
})

```

```

df = pd.DataFrame(results)
df = df.sort_values(by="Final AI Fit Score", ascending=False)

```

df

<div>

	Name	Semantic Score	Skill Score	Experience Score	Final AI Fit Score
0	John Doe	0.859594	0.70	0.3	0.699797
2	Priya Singh	0.636923	0.40	0.1	0.458461
3	Arjun Mehta	0.534216	0.35	0.4	0.452108
1	Rahul Sharma	0.635388	0.25	0.2	0.432694
4	Kiran Patel	0.401513	0.00	0.5	0.300756

<div class="colab-df-buttons">

<button class="colab-df-convert" onclick="convertToInteractive('df-3d84d433-8' title="Convert this dataframe to an interactive table." style="display:none;">

```
<style>
.colab-df-generate {
  background-color: #E8F0FE;
  border: none;
  border-radius: 50%;
  cursor: pointer;
  display: none;
  fill: #1967D2;
  height: 32px;
  padding: 0 0 0 0;
  width: 32px;
}

.colab-df-generate:hover {
  background-color: #E2EBFA;
  box-shadow: 0px 1px 2px rgba(60, 64, 67, 0.3), 0px 1px 3px 1px rgba(60, 64, 67, 0.3);
  fill: #174EA6;
}

[theme=dark] .colab-df-generate {
  background-color: #3B4455;
  fill: #D2E3FC;
}

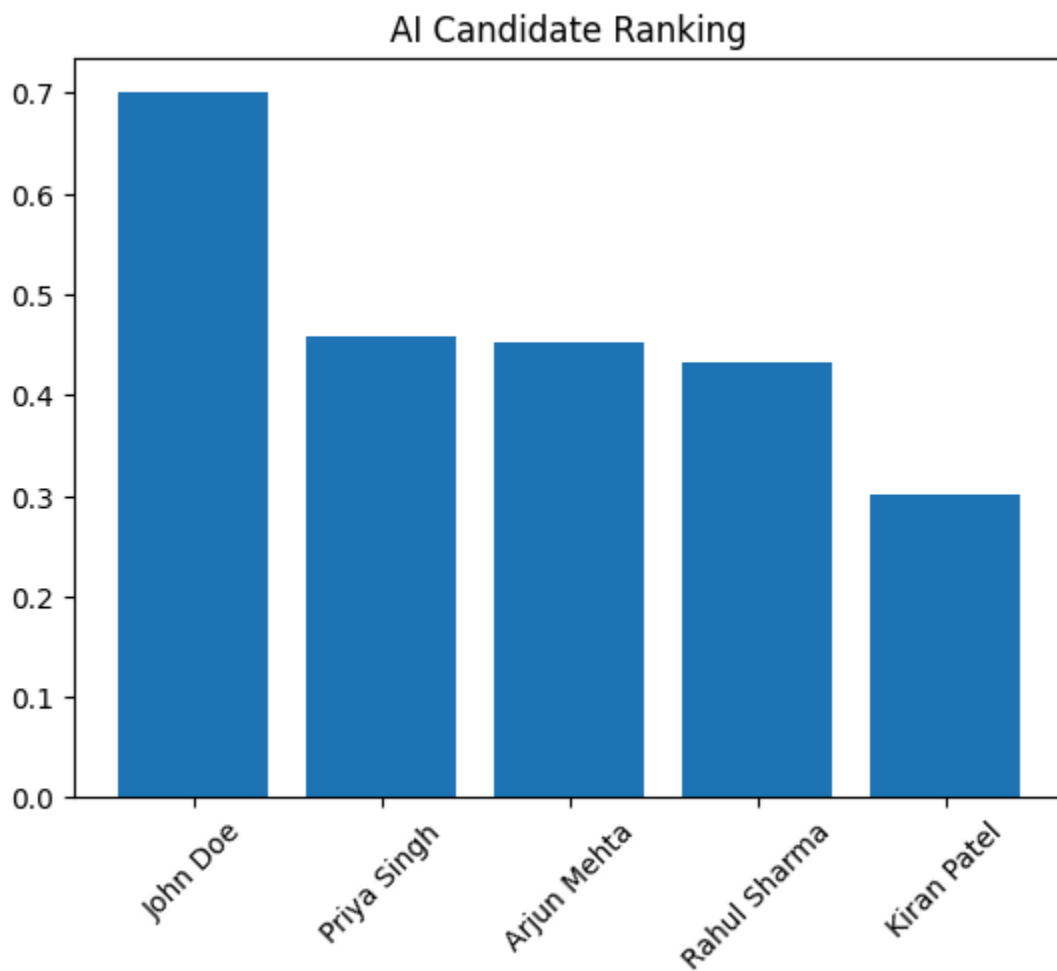
[theme=dark] .colab-df-generate:hover {
  background-color: #434B5C;
  box-shadow: 0px 1px 3px 1px rgba(0, 0, 0, 0.15);
  filter: drop-shadow(0px 1px 2px rgba(0, 0, 0, 0.3));
  fill: #FFFFFF;
}
```

```
</style>  
<button class="colab-df-generate" onclick="generateWithVariable('df')"  
  title="Generate code using this dataframe."  
  style="display:none;">
```



```
</div>
```

```
import matplotlib.pyplot as plt  
  
plt.figure()  
plt.bar(df["Name"], df["Final AI Fit Score"])  
plt.xticks(rotation=45)  
plt.title("AI Candidate Ranking")  
plt.show()
```



png